

Designing Low Noise Amplifier (LNA) using Open-Source Tools

Rachit Saini¹ and Alok Kumar Saxena²

¹Department of Electrical & Instrumentation Engineering, Thapar Institute of Engineering & Technology, Patiala, India

²Department of Electrical Engineering, Indian Institute of Technology Jammu, India

Abstract

Low Noise Amplifiers (LNAs) operating in the D-band (110–170 GHz) are critical components in 5G, radar, and imaging systems. Commercial EDA tools for such designs are prohibitively expensive, restricting academic research. This work presents the design, simulation, and electromagnetic (EM) verification of a 130 GHz cascode LNA using exclusively open-source tools with the IHP SG13G2 130 nm SiGe BiCMOS PDK. Qucs-S with Ngspice is used for schematic entry and S-parameter simulation; KLayout for physical layout; and openEMS for 3D FDTD EM characterization of a Differential Stacked Spiral Inductor (DSSI).

1. Introduction

LNAs are the first gain stage in receiver chains, making noise figure and gain fundamental to system performance. Designing at these frequencies requires high-fidelity HBT transistor models and EM-characterised passives, historically locked behind expensive commercial licenses. The public release of the IHP SG13G2 PDK (130 nm SiGe BiCMOS; $f_T/f_{max} = 350/450$ GHz) removes the technology barrier (Herman et al., 2024). However, unlike digital IC design where OpenROAD provides a unified open pipeline (Chupilko et al., 2021), analog/RF design still lacks a coherent open toolchain. This project fills that gap by independently validating and extending the open-source RFIC flow of Simone et al. (2025).

2. Methodology

The flow (Figure 1a) has four stages. (1) **Schematic (Qucs-S)**: Cascode LNA with npn13G2 HBTs (Q1, Q2; $8 \times 0.9 \mu\text{m}$, $N_x=8$) and current-mirror bias (Q3). Input matching: $L1 = 42 \text{ pH}$, $C2 = 38 \text{ fF}$, $L2 = 100 \text{ pH}$. Output: $L3 = 10 \text{ nH}$, $L4 = 117 \text{ pH}$, $C6 = 31 \text{ fF}$. $VCC = 2.2575 \text{ V}$; PDK models from cornerHBT.lib (hbt_typ). (2) **Simulation (Ngspice)**: S-parameters over 110–170 GHz; NF/NF_{min} extracted natively; μ and μ' via Nutmeg from S-matrix determinant. (3) **Layout (KLayout)**: DSSI designed via gdstk scripting with IHP SG13G2 layer rules. (4) **EM Simulation (openEMS)**: 3D FDTD simulation of DSSI; Touchstone (.s1p) re-imported into Qucs-S for co-simulation.

3. Results / Expected Outcome

At 130 GHz (Table 1), $S_{21} \approx 12.1 \text{ dB}$ matches the reference (12.2 dB, $\Delta = 0.1 \text{ dB}$). $S_{11} \approx -20.1 \text{ dB}$ and $S_{22} \approx -19.8 \text{ dB}$ compare reasonably with reference values of -24.1 and -22.4 dB ; small offsets are attributed to empirical bias tuning. $NF \approx 5.5 \text{ dB}$ matches the reference (5.2 dB, $\Delta = 0.3 \text{ dB}$). Stability factors $\mu > 1.5$ and $\mu' > 1.4$ confirm unconditional stability. DSSI EM simulation yielded $L = 2.73 \text{ nH}$ and $Q \approx 0.19$ at 1 GHz, capturing substrate loss not modelled by ideal inductors. This EM characterisation of the input DSSI is the novel extension beyond Simone et al. (2025), which simulated only the output transformer.

Keywords: Low-noise amplifier, open-source EDA, SiGe BiCMOS, millimeter-wave, EM simulation, IHP SG13G2 PDK.

Figures / Illustrations

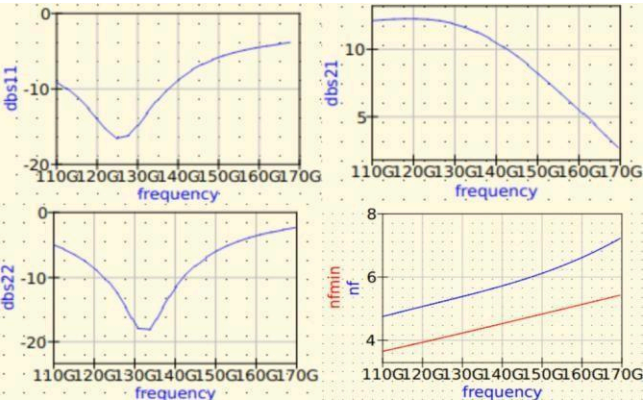


Figure 1. S-parameters & Noise Figure of the cascode LNA schematic in Qucs-s.

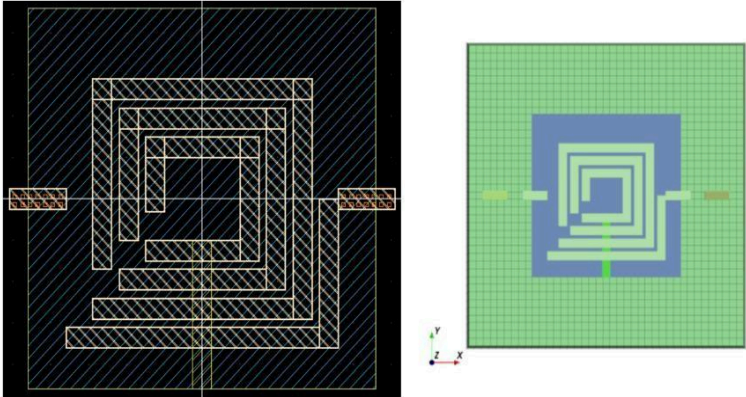


Figure 2. DSSI spiral inductor EM characterisation - Physical layout in KLayout with IHP SG13G2 metal layers & 3D FDTD model in openEMS.

References 1. Brinson, M., & Salfelder, F. (2021). Compact device modeling with Qucs/Qucs-S/Xyce. In Proc. 28th MIXDES (pp. 35–40). Łódź. 2. Chupilko, M., et al. (2021). Survey of open-source flows for digital hardware design. In 2021 IVMEM (pp. 11–16). Moscow. 3. Herman, K., Scholz, R., & Andreev, S. (2024). Reflections on the first European open-source PDK by IHP. In 2024 MIXDES (pp. 19–22). 4. Simone, G., et al. (2025). Validation of open-source RFIC design flow using 130 GHz LNA. In 2025 APMC. https://doi.org/10.1109/APMC65046.2025.11378430 <td> Student Profile  Name: Rachit Saini Institute: Thapar Institute of Engineering & Technology, Patiala, India RISE-UP Batch: 2026 (May–July) [TID-2026-216] Guide: Dr. Alok Kumar Saxena, Assistant Professor, Department of Electrical Engineering, IIT Jammu Alok Saxena [Supervisor’s Signature] </td>	Student Profile  Name: Rachit Saini Institute: Thapar Institute of Engineering & Technology, Patiala, India RISE-UP Batch: 2026 (May–July) [TID-2026-216] Guide: Dr. Alok Kumar Saxena, Assistant Professor, Department of Electrical Engineering, IIT Jammu Alok Saxena [Supervisor’s Signature]
--	--